8 ta Agentic AI Loyha - Qisqa Tavsif + Streamlit UI

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1. 📊 Sales Data Analyzer Agent

Qisqa tavsif: CSV/Excel satish ma'lumotlarini o'qib, analiz qiluvchi va user savollariga javob beradigan sodda agent. Foydalanuvchi faylni upload qila oladi, ma'lumotlarni jadval ko'rinishida ko'ra oladi va natural language savollar orqali statistika, top produktlar, regional analiz kabi ma'lumotlarni olishi mumkin. Streamlit'da simple metrics, bar chartlar va dataframe displayda ko'rsatiladi.

Streamlit UI Components: File uploader, dataframe view, metric cards, bar chart, text input, response display

Talablar:

- ✅ Streamlit app'i Streamlit Cloud'da o'rnatilgan bo'lishi kerak
- ✅ Live URL orqali browser'da ochilishi kerak
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2. 🤖 Smart FAQ Chatbot (RAG)

Qisqa tavsif: Company FAQ va Knowledge Base'ni vectorize qilip, customer savollariga to'g'ri javob beradigan AI chatbot. Admin tomondan FAQ dokumentlarini upload qila oladi, semantik search orqali relevant dokumentlarni topadi va LLM'dan javob yaratadi. Chat interface'da user va assistant messages ketma-ketda ko'rsatiladi. Sidebar'da FAQ upload va conversation history mavjud.

Streamlit UI Components: Chat message interface, text input, file uploader, sidebar history, progress bar, session state

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3. 🎯 Lead Scoring & Pipeline Manager

Qisqa tavsif: CRM lead'larni tahlil qilip, conversion ehtimoli hisoblaydi va sales strategy tavsiya qiladi. Admin CSV orqali leads o'z ichiga olgan ma'lumotlarni yuklaydi, har bir lead uchun tahlil,

conversion probability va pricing recommendation ko'rsatiladi. Sidebar'da budget, industry, conversion probability bo'yicha filterlar bor. Columns'da KPI metriclar (high priority leads, conversion rate, total value), dataframe jadvalda leads va charts'da lead distribution ko'rsatiladi.

Streamlit UI Components: Multiple columns for KPI display, dataframe with sorting, sidebar filters (slider, multiselect), line/bar charts, expandable details

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4. 📄 Intelligent Invoice & Receipt OCR Agent

Qisqa tavsif: PDF/Image'lardan invoice, receipt va barcode'larni o'qib, ma'lumotlarni automatic extract qiladi. Foydalanuvchi PDF yoki image upload qiladi, preview ko'rsatiladi va extracted data'lar (invoice number, amount, date, vendor, emails) JSON ko'rinishida va formatted fields'da ko'rsatiladi. Success message'si bilan database'ga save qilindi haqida inform qiladi.

Streamlit UI Components: File uploader (PDF/Image), image display, JSON output, metric cards, success/error messages, database save confirmation

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5. 🎨 Natural Language to Dashboard Generator

Qisqa tavsif: Natural language'dan (masalan "Top 5 muhimning shu oyning savdosu") automatic interactive dashboard generate qiladi. User ingiz tilida savol yozadi, system Text-to-SQL orqali SQL query yaratadi, optimal chart type'ni tanlaydi va dashboard render qiladi. Generated SQL code'i ko'rsatiladi, multiple charts va raw data dataframe'da ko'rsatiladi. CSV va PDF format'da download qilish imkoniyati mavjud.

Streamlit UI Components: Text area for natural language input, code block for SQL display, multiple plotly charts, dataframe view, download buttons, query results display

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6. 💹 Real-Time Market Monitoring Agent

Qisqa tavsif: Stock prices, news, economic indicators'ni real-time monitoring qilip, anomalies aniqlaydi va alerts beradi. Foydalanuvchi symbols multiselect qiladi (AAPL, GOOGL, MSFT, TSLA), real-time price metrics ko'rsatiladi. Real-time chart auto-refresh bo'ladi har 30 sekundda. Active alerts jadvalda ko'rsatiladi, market news'lar sentiment bilan birga ko'rsatiladi va economic indicators'ni dataframe'da ko'rish mumkin.

Streamlit UI Components: Multiselect for symbol selection, metric cards for real-time prices, auto-refresh with st.rerun(), real-time chart, alerts dataframe, news sentiment display, economic indicators table

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7. 🏢 Multi-Agent Business Intelligence System

Qisqa tavsif: Bir nechta specialized agent'larni bir platforma'da orchestrate qilip, complex business queries handle qiladi. Sidebar'da Dashboard, Sales Analysis, Inventory, Finance, HR Analytics pages mavjud. Har bir page uchun o'ziga xos UI bor - metrics, charts, agent responses. Main dashboard'da jami KPI'lar (Total Revenue, Active Deals, Inventory Value, Employee Count) ko'rsatiladi. Bottom'da hammasiga umumiy chat interface bor - user har qanday savol qo'yishi mumkin va agent orchestrator javob beradi.

Streamlit UI Components: Sidebar radio navigation, multiple page layouts, KPI metrics columns, various charts per page, chat interface at bottom, session state for multi-page navigation

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8. 🔐 Enterprise Contract Analyzer & Risk Agent

Qisqa tavsif: Company shartnomalarni o'qib, key terms, risks, obligations'ni aniqlaydi va legal alerts beradi. Admin shartnoma PDF'ni upload qiladi, left column'da preview, right column'da risk score bar va metric cards (Key Terms Found, Red Flags, Critical Dates) ko'rsatiladi. Tabs'da Overview, Risk Analysis, Key Terms va Q&A mavjud. Risk Analysis'da red flags va critical items warning/error messages bilan ko'rsatiladi. Q&A tab'da user contract haqida savollar qo'ya oladi. Oxirida PDF va JSON report'larini download qilish mumkin.

Streamlit UI Components: File uploader, two-column layout, progress bar for risk score, metric cards, tabs for different sections, warning/error/info messages, dataframe for risks and terms, text input for Q&A, download buttons

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📊 Barcha Loyhalari Taqqoslash (Streamlit UI bilan)

#	Nomi	UI Complexity	Business Use
1	Sales Data Analyzer	Simple	Sales Analytics
2	FAQ Chatbot (RAG)	Medium	Customer Support
3	Lead Scoring	Medium	Sales CRM
4	Invoice OCR	Medium	Accounting
5	NL to Dashboard	Advanced	Business Analytics
6	Market Monitor	Advanced	Trading/Finance
7	Multi-Agent BI	Complex	Enterprise BI
8	Contract Analyzer	Advanced	Legal/Compliance

🎨 Streamlit Features Used

Simple Features (Loyha 1-3):

- Dataframe tables
- Metric cards
- Bar charts va line charts
- Text input va text area
- Buttons
- Session state

Medium Features (Loyha 4-5):

- File uploader
- Two/three column layouts
- Tabs navigation
- Expandable sections (expander)
- Plotly charts
- Download buttons
- JSON display

Advanced Features (Loyha 6-8):

- Page config customization
- Sidebar navigation
- Radio buttons va multiselect
- Chat message interface
- Progress bars
- Alert messages (warning, error, success, info)
- Auto-refresh (st.rerun)
- Session state management
- Real-time updates

🎓 O'rganish Tavsiya Etilgan Tartib

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1️⃣ Sales Data Analyzer (Streamlit asoslar - dataframe, charts, metrics)
   ↓
2️⃣ FAQ Chatbot (Chat UI + session state)
   ↓
3️⃣ Lead Scoring (Filters + multi-column layout)
   ↓
4️⃣ Invoice OCR (File upload + image display)
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    ↓
5️⃣ NL to Dashboard (Tabs + advanced charts)
    ↓
6️⃣ Market Monitor (Real-time + auto-refresh)
    ↓
7️⃣ Multi-Agent BI (Sidebar navigation + multi-page)
    ↓
8️⃣ Contract Analyzer (Complex layout + tabs + chat)
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💡 Har Bir Loyha uchun Stack

Loyha 1: Sales Data Analyzer

- Frontend: Streamlit
- Backend: Pandas, Python
- Database: SQLite/PostgreSQL (optional)

Loyha 2: Smart FAQ Chatbot

- Frontend: Streamlit
- Backend: LangChain, LLM API (Claude/GPT-4)
- Vector DB: Pinecone, Weaviate, FAISS
- Embedding Model: OpenAI, HuggingFace

Loyha 3: Lead Scoring

- Frontend: Streamlit
- Backend: Pandas, scikit-learn
- Database: PostgreSQL
- Prediction: Machine Learning model

Loyha 4: Invoice OCR

- Frontend: Streamlit
- Backend: PyPDF2, python-docx, spaCy, OpenCV
- NLP: Regex, Named Entity Recognition
- Database: PostgreSQL, SQLAlchemy ORM

Loyha 5: NL to Dashboard

- Frontend: Streamlit

- Backend: LangChain, LLM API
- Database: PostgreSQL, BigQuery
- Text-to-SQL: LLM-based query generation

Loyha 6: Real-Time Market Monitor

- Frontend: Streamlit
- Backend: AsyncIO, APIs (Alpha Vantage, NewsAPI)
- Message Queue: Kafka (optional)
- Real-time Processing: pandas, numpy
- LLM: Claude/GPT-4 for analysis

Loyha 7: Multi-Agent BI

- Frontend: Streamlit
- Backend: LangChain, LlamaIndex
- API: FastAPI
- Database: PostgreSQL, Data Warehouse
- Orchestration: ReAct pattern, Tool calling

Loyha 8: Contract Analyzer

- Frontend: Streamlit
- Backend: LangChain, LLM API
- NLP: spaCy, Named Entity Recognition
- Vector DB: FAISS, Pinecone (RAG uchun)
- PDF Processing: PyPDF2, pdfplumber